

Product Requirements Document

Product Name: PalmInsight (working name)

Product Type: AI-powered palm reading web application

Platform: Web (mobile-first)

1. Product Vision

Build an interactive web application where users upload or capture a photo of their palm, and the system automatically:

Detects palm lines

Highlights them graphically

Generates interpretations based on palmistry rules

Offers deeper insights via a paid upgrade.

The experience should feel visual, interactive, and shareable, similar to a personality test or astrology reading.

2. Target Users

Primary

Gen Z (18–30)

Social media users

Astrology / personality enthusiasts

Secondary

Casual curiosity users

Friends comparing palm readings

Content creators sharing results

3. Core Value Proposition

Users can:

Upload a hand photo

See AI-detected palm lines

Get instant interpretations

Unlock deeper analysis through premium reports

4. User Flow

Step 1 – Landing Page

User sees:

Hero: “Scan Your Palm. Discover Your Story.”

CTA: Scan Palm

Step 2 – Capture / Upload

User can:

Upload image

Use camera (mobile)

Requirements:

Palm clearly visible

Good lighting

Open hand

System performs:

Hand detection

Palm segmentation

Step 3 – AI Analysis

Pipeline:

Image Upload

↓

Hand Detection

↓

Palm Region Extraction

↓

Line Detection

↓

Palmistry Rule Engine

↓

Result Generation

5. Palm Lines to Detect

Life Line

Location: Curves around thumb.

Used for:

energy

resilience

lifestyle tendencies

Interpretation examples:

Pattern Meaning

Deep strong vitality

Long stable life path

Broken major life changes

Heart Line

Location: Top horizontal line.

Used for:

emotional nature

love tendencies

Examples:

Pattern Meaning

Curved romantic nature

Straight logical approach to love

Multiple branches emotional complexity

Head Line

Location: middle horizontal line.

Used for:

thinking style

decision making

Examples:

Pattern Meaning

Long analytical mind

Curved creative thinking

Straight practical mindset

Fate Line

Location: vertical line toward middle finger.

Used for:

career tendencies

life direction

Examples:

Pattern Meaning

Deep strong career focus

Faint flexible career path

Broken career shifts

6. Visual Result Interface

The system overlays colored line highlights.

Example UI behavior:

Palm Image Displayed

Life Line → highlighted in red

Heart Line → highlighted in blue

Head Line → highlighted in yellow

Fate Line → highlighted in green

When the user taps a line:

Panel expands with explanation.

7. Result Components

Each line generates:

Card Layout

Title

Line type

Detected pattern

Example: "Long & Deep"

Interpretation text

Confidence level

Example: 82%

Visual overlay toggle

8. Freemium Model

Free Tier

Users receive basic reading.

Free features:

Life line analysis

Heart line analysis

Includes:

visual highlight

short explanation

Example result:

Life Line

Detected: Deep and Long

Meaning:

You are likely resilient and able to recover from challenges.

You tend to maintain strong personal energy levels.

Paid Tier (\$5)

Unlock Deep Report

Includes:

Head Line analysis

Fate Line analysis

Additional sections:

Full destiny report

Love compatibility analysis

Career tendency report

Example premium output:

Career Tendency

Your fate line suggests gradual career progress.

You may experience a key turning point in your late 20s.

Best-suited environments:

Entrepreneurship

Technology

Strategic roles

9. Love Compatibility Feature (Premium)

User can:

Upload two palms

or

enter partner birth date.

System generates:

Compatibility score

Example:

Compatibility Score: 74%

Strengths:

Strong emotional understanding

Potential challenges:

Different decision styles

10. Career Insight Feature

Based on:

fate line

head line

palm structure

Possible outputs:

Entrepreneurial tendency

Creative professions

Strategic leadership

Technical fields

11. Shareability

Users can generate:

Instagram story card

share link

Example:

“AI says my palm shows strong leadership potential.”

This helps viral growth.

12. AI Architecture

Image Processing

Possible libraries:

OpenCV

MediaPipe Hands

Steps:

- 1 Detect hand landmarks
- 2 Crop palm
- 3 Detect edges
- 4 Identify line candidates

Rule Engine

Example logic:

if life_line_length > threshold
 meaning = strong vitality

if heart_line_curved
 meaning = romantic personality

Text Generation

Templates or LLM generated explanations.

13. Database Requirements

Store:

User session
Uploaded images (optional)
Result metadata
Payment status
Premium unlock status

Example schema:

Users
PalmScans
Reports
Payments

14. Tech Stack

Frontend

Next.js

React
Canvas / SVG overlays

Backend

Node.js
Express API

Image Processing

OpenCV / MediaPipe

Payments

Stripe

15. Database Choice
Option A – MongoDB

Pros:

Flexible schema
Good for storing JSON results
Large ecosystem

Cons:

Need hosting setup.

Option B – Supabase

Pros:

Auth built in
Realtime database
Easy storage

Cons:

Relational schema slightly rigid.

Recommendation

For this product:

Supabase is better for MVP.

Reasons:

Built-in auth

Storage for images

Simple API

Faster development

MongoDB is better later if scaling.

16. Privacy Requirements

Users must be informed:

Palm images are analyzed automatically.
Images may be deleted after processing.

17. MVP Scope

Initial release should include:

Palm upload
Line detection
Life line result
Heart line result
Premium paywall
Share card

Everything else can come later.

18. Monetization Goal

Conversion target:

3–5% users upgrade.

Example:

10k users

→ 300 paid

Revenue:

$$300 \times \$5 = \$1500$$

Final Summary

The product combines:

Computer vision
Personality insight
Shareable content

With a freemium model:

Free:

Life line
Heart line

Paid (\$5):

Full destiny report
Love compatibility
Career prediction